

1.7 The Circular Cylindrical Coordinate System

Some important features of circular cylindrical coordinate system

- a) It is the three-dimensional version of the polar coordinates of analytic geometry.
- b) We specify the distance z of the point from an arbitrary $z = 0$ reference plane that is perpendicular to the line $\rho = 0$.
- c) We consider any point as the intersection of three mutually perpendicular surfaces. These surfaces are a circular cylinder ($\rho = \text{constant}$), a plane ($\phi = \text{constant}$), and another plane ($z = \text{constant}$). Note that three such surfaces may be passed through any point, unless it lies on the z axis, in which case one plane suffices.

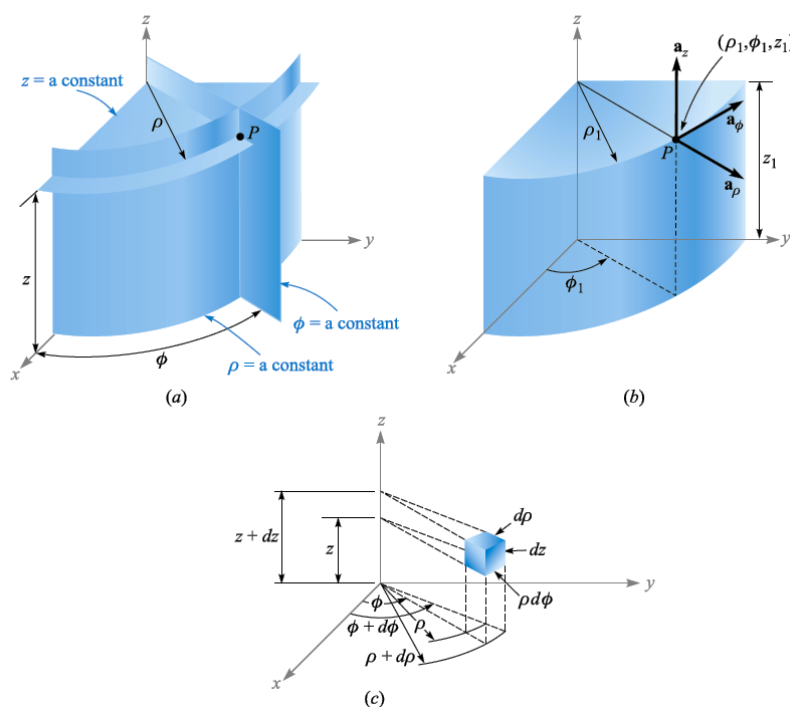


Fig. 1.8 (a) The three mutually perpendicular surfaces of the circular cylindrical coordinate system. (b) The three unit vectors of the circular cylindrical coordinate system. (c) The differential volume unit in the circular cylindrical coordinate system; $d\rho$, $\rho d\phi$, and dz are all elements of length.

d) The unit vectors in this case are directed towards increasing coordinate values and are perpendicular to the surface on which that coordinate value is constant (i.e., the unit vector $\mathbf{a}_x$ is normal to the plane $x = \text{constant}$ and points toward larger values of x). In a corresponding way, we may now define three unit vectors in cylindrical coordinates, $\mathbf{a}_\rho$, $\mathbf{a}_\phi$, and $\mathbf{a}_z$.

e) The unit vector $\mathbf{a}_\rho$ at a point $P(\rho_1, \phi_1, z_1)$ is directed radially outward, normal to the cylindrical surface $\rho = \rho_1$. It lies in the planes $\phi = \phi_1$ and $z = z_1$. The unit vector $\mathbf{a}_\phi$ is

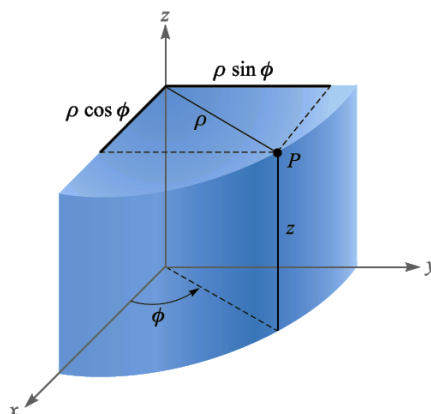


Fig. 1.9 The relationship between the rectangular variables x, y, z and the cylindrical coordinate variables ρ, ϕ, z . There is no change in the variable z between the two systems.

normal to the plane $\phi = \phi_1$, points in the direction of increasing ϕ , lies in the plane $z = z_1$, and is tangent to the cylindrical surface $\rho = \rho_1$. Fig. 1.8(b) shows the three vectors in cylindrical coordinates.

f) In rectangular coordinates, the unit vectors are not functions of the coordinates. Two of the unit vectors in cylindrical coordinates, $\mathbf{a}_\rho$ and $\mathbf{a}_\phi$, however, do vary with the coordinate ϕ , as their directions change. In integration or differentiation with respect to ϕ , $\mathbf{a}_\rho$ and $\mathbf{a}_\phi$ must not be treated as constants.

g) $\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$.

h) A differential volume element in cylindrical coordinates may be obtained by increasing ρ, ϕ , and z by the differential increments $d\rho, d\phi$, and dz .

- The two cylinders of radius ρ and $\rho + d\rho$, the two radial planes at angles ϕ and $\phi + d\phi$, and the two "horizontal" planes at "elevations" z and $z + dz$ now enclose a small volume, as shown in Figure 1.8(c), having the shape of a truncated wedge.
- As the volume element becomes very small, its shape approaches that of a rectangular parallelepiped having sides of length $d\rho$, $\rho d\phi$, and dz .
- Note that $d\rho$ and dz are dimensionally lengths, but $d\phi$ is not; $\rho d\phi$ is the length. The surfaces have areas of $\rho d\rho d\phi$, ρdz , $\rho d\phi dz$, and the volume becomes $\rho d\rho d\phi dz$.

The variables of the rectangular and cylindrical coordinate systems are easily related to each other. Referring to Figure 1.9, we see that

$$\begin{aligned}x &= \rho \cos \phi \\y &= \rho \sin \phi \\z &= z\end{aligned}$$

From the other viewpoint, we may express the cylindrical variables in terms of x, y , and z :

$$\begin{aligned}\rho &= \sqrt{x^2 + y^2} \quad (\rho \geq 0) \\ \phi &= \tan^{-1} \frac{y}{x} \\ z &= z\end{aligned}$$

Transformation of Vectors: Table Dot products of unit vectors in cylindrical and rectangular coordinate systems

	$\mathbf{a}_\rho$	$\mathbf{a}_\phi$	$\mathbf{a}_z$
$\mathbf{a}_x \cdot$	$\cos \phi$	$-\sin \phi$	0
$\mathbf{a}_y \cdot$	$\sin \phi$	$\cos \phi$	0
$\mathbf{a}_z \cdot$	0	0	1

Example: Transform the vector $\mathbf{B} = y\mathbf{a}_x - x\mathbf{a}_y + z\mathbf{a}_z$ into cylindrical coordinates.

Solution: The new components are

$$\begin{aligned}B_\rho &= \mathbf{B} \cdot \mathbf{a}_\rho = y(\mathbf{a}_x \cdot \mathbf{a}_\rho) - x(\mathbf{a}_y \cdot \mathbf{a}_\rho) \\ &= y \cos \phi - x \sin \phi = \rho \sin \phi \cos \phi - \rho \cos \phi \sin \phi = 0 \\ B_\phi &= \mathbf{B} \cdot \mathbf{a}_\phi = y(\mathbf{a}_x \cdot \mathbf{a}_\phi) - x(\mathbf{a}_y \cdot \mathbf{a}_\phi) \\ &= -y \sin \phi - x \cos \phi = -\rho \sin^2 \phi - \rho \cos^2 \phi = -\rho \\ B_z &= \mathbf{B} \cdot \mathbf{a}_z = z\end{aligned}$$

Thus,

$$\mathbf{B} = -\rho \mathbf{a}_\phi + z \mathbf{a}_z$$